Royal Devon University Healthcare NHS Foundation Trust

Administration of T-VEC (Talimogene Laherparepvec or Imlygic) for the Treatment of Malignant Melanoma in Oncology Outpatients	
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Note: This document has been assessed for any equality, diversity or human rights implications			

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1	06/08/2018	Immunotherapy CNS	New document, to meet requirements for implementation of T-VEC outpatient clinic.
1.1	20/08/2019	Immunotherapy CNS	Minor amendments throughout and addition of information regarding admission of a patient receiving T-VEC.
2	01/05/2020	Oncology Lead	Routine review. Skin CNS's removed from administrators (8.6) and requirement to cover bag from light once drawn up. Maximum storage times amended (7.2.1).
3	04/05/2022	Immunotherapy CNS	Routine review. Updates to administrators and prescribers. Inclusion of Epic (MYCARE).

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CONTENTS

1. INTRODUCTION 5
2. PURPOSE 5
3. DUTIES AND RESPONSIBILITIES OF STAFF 6
4. SERVICE DELIVERY 6
5. DOCUMENTATION 6
6. ENVIRONMENTAL ASPECTS 6
7. STABILITY OF T-VEC 7
8. TREATMENT ADMINISTRATION 8
9. ADMISSION OF PATIENT RECEIVING T-VEC 9
10. ARCHIVING ARRANGEMENTS 10
11. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE STANDARD OPERATING PROCEDURE 10
12. REFERENCES 10
13. ACKNOWLEDGEMENTS 10

APPENDIX 1: RECOMMENDED DOSE AND SCHEDULE FOR T-VEC ADMINISTRATION 11

APPENDIX 2: CLOSE CONTACTS INFORMATION 12

KEY POINTS OF THIS PROCEDURAL DOCUMENT:

- Service delivery
- Documentation
- Environmental aspects
- Stability of T-VEC
    - Thawing T-VEC vials
    - After thawing
    - Process
- Treatment administration
- Admission of a patient receiving T-VEC

1. INTRODUCTION

1.1 Talimogene Laherparepvec (T-VEC) is an oncolytic virus recently approved and licensed for the treatment of malignant melanoma at specific stages.

1.2 NICE has approved T-VEC for the treatment of metastatic melanoma involving skin and lymph nodes without evidence of visceral, brain or bone disease. This service has been developed at the Royal Devon and Exeter NHS Foundation Trust (hereafter referred to as "the Trust") in order to deliver T-VEC to suitable patients in a specialist clinic.

1.3 The service will be led by Dr Kate Scatchard, Consultant in Medical Oncology.

2. PURPOSE

2.1 The licensed indication for T-VEC approved by NICE is for adult patients with un-resectable melanoma that is regionally or distantly metastatic, i.e. skin lesions and nodal spread (stage IIIb, IIIc and stage IV M1a), with no brain, bone, lung or other visceral disease.

2.2 T-VEC is a live virus which has been specially engineered for the treatment of cancer by direct intra-tumoural injection. The licensing of the use of T-VEC in malignant melanoma occurred in October 2015. The treatment was approved by NICE in September 2016 (TA 410).

2.3 T-VEC is a cancer killing virus (termed an oncolytic virus) which can kill cancer cells upon direct injection. The destruction of cancer cells at the injection site stimulates an anti-tumour immune response which can spread to cancer lesions at other sites remote from the injection site.

2.4 T-VEC progressed through phase I and phase II studies and ultimately the regulatory approval for its use was based on the publication of the phase III study called OPTiM. This study showed clear evidence of significant disease control with repeated intra-tumoural injections of T-VEC.

2.5 The virus is based on a herpes simplex type 1 cold sore virus. The agent has been carefully engineered to eliminate typical virus associated pathology such as infection and latency within the nervous system. It is safe for patients, relatives and the environment. It is a licenced and approved agent for use in hospital, provided there facilities exist for storage and delivery to patients.

2.6 T-VEC can kill cancer cells but spares normal cells. Treatment is given by intra-tumoural injection by trained staff. The virus can infect both cancer cells and normal cells but is only able to replicate in cancer cells. The virus is armed with an immuno-stimulatory molecule called GMCSF. This enhances immune activation and aids the generation of a whole-body immune response against non-injected lesions.

2.7 In the OPTiM study [2], 430 patients with unresectable stage IIIb, IIIc or stage IV M1a melanoma were treated in a 2:1 randomisation. The primary endpoint was durable response rate (DRR) defined as a complete or partial tumour response lasting at least 6 months and starting within 12 months of treatment. The durable response rate was 25% with T-VEC vs 1.2% in the control arm (patients injected only with an immune stimulating protein, GMCSF). The median overall survival for the T-VEC treated patients was 41.1 months vs 21.5 months with the GMCSF only arm.

2.8 T-VEC has been well tolerated, even in older patients with lower performance status. The most commonly encountered side-effects are occasional chills, fever and fatigue.

3. DUTIES AND RESPONSIBILITIES OF STAFF

3.1 Suitable patients will be identified by Dr Scatchard.

3.2 Operators who will administer T-VEC are currently Dr Scatchard, Sally Tapp Lead Immunotherapy CNS and Immunotherapy Clinical Nurse Specialists who have undergone training.

4. SERVICE DELIVERY

4.1 T-VEC will be delivered to patients with malignant melanoma as outlined by the licenced indication.

4.2 This will take place at in Oncology Outpatient's, Royal Devon and Exeter NHS Trust.

4.3 Administration guidelines provided by the parent company AMGEN will be closely adhered to.

4.4 Specific referral will be by a clinician, usually through the skin cancer MDT and the aim is to see the patient within 2 weeks.

4.5 Patients will have baseline documentation of disease burden by means of medical photography and marker lesion measurements. A baseline CT or PET scan will also be done and repeated at three-monthly intervals to monitor changes in disease burden.

4.6 Treatment will be administered based on doses and instructions issued by the manufacturer (Appendix 1).

4.7 Clear evidence of significant disease progression will trigger a referral of the patient back to the referring clinician for discussion re treatment management.

4.8 There is no indication that treatment with T-VEC negatively affects the likelihood of subsequent response to other treatments for malignant melanoma including BRAF inhibitors and Immunotherapies such as Ipilimumab, Pembrolizumab, Nivolumab etc.

5. DOCUMENTATION

5.1 The record sheets will be used to document a course of treatment for each patient.

5.2 T-VEC treatment records for all patients receiving this treatment will be stored on EPIC.

6. ENVIRONMENTAL ASPECTS

6.1 The clinic rooms 4 or 5 in oncology outpatients are an appropriate facility for administration of this treatment with booked slots of 1 hour.

6.2 All required safety equipment and protective clothing will be available. AMGEN provide a patient information booklet for patients who have been offered and are considering treatment with T-VEC.

6.3 Specific information will be provided to the patient's GP and practice/community nurse, if required, regarding the treatment, schedule, dressings and aftercare, side effects, and contact details.

6.4 Close contacts will be provided with specific information regarding Imlygic treatment and signs and symptoms of Herpes infection. (Appendix 2)

7. STABILITY OF T-VEC

7.1 Thawing T-VEC vials

7.1.1 T-VEC vials must be thawed until the entire contents are in a liquid state before giving. This process takes approximately 30 minutes at ambient room temperature (20-25°C).

7.1.2 Vials should be gently swirled and not shaken.

7.1.3 Vials should be thawed and stored in the original carton until administration in order to protect from light.

7.2 After thawing

7.2.1 After thawing, administer T-VEC as soon as practically feasible.

Thawed T-VEC stability when stored at temperatures of 2°C up to 25°C protected from light in its original vial, in a syringe, or in the original vial followed by a syringe is according to table below. Do not exceed the storage times specified.

Maximum storage time for thawed T-VEC in a syringe	10^6 (1 million) PFU/mL	10^8 (100 million) PFU/mL
2ºC to 8ºC	8 hours	8 hours
up to 25ºC	2 hours	4 hours

Maximum storage time (storage time in vial + storage time in syringe) for thawed T-VEC	10^6 (1 million) PFU/mL	10^8 (100 million) PFU/mL
2ºC to 8ºC	24 hours	1 week (7 days)
up to 25ºC	12 hours	24 hours

7.2.2 T-VEC must not be refrozen once it has thawed. Discard any thawed T-VEC in the vial or syringe stored longer than the specified times.

7.3 Process

7.3.1 Patient identified at MDT.

7.3.2 Clinic appointment to assess suitability/discuss treatment/provide written information/consent.

7.3.3 Baseline CT or PET will be performed.

7.3.4 Baseline assessments: Lesion measurement, photography (CNS to measure, take consent for photographs and photograph lesions on treatment day 1).

7.3.5 2nd treatment at 3 week interval after initial treatment.

7.3.6 Subsequent treatments (i.e. cycle 3 onwards) at 2 weekly intervals.

7.3.7 Photographs of all lesions (including new lesions) at each treatment visit. Document measurements on treatment record sheets.

7.3.8 Blood tests (FBC, U+E, LFT, calcium, LDH) at baseline and prior to alternate treatment.

7.3.9 CT or PET every 12 weeks.

7.3.10 Continue treatment for at least 6 months unless significant disease progression requires consideration of alternative treatment.

8. TREATMENT ADMINISTRATION

8.1 Suitable patients will be identified by Dr K Scatchard ; treatments will only be prescribed by Dr K Scatchard /Sally Tapp. A baseline record of lesions to be injected will be made and documented on EPIC in terms of location, number, size and whether to be injected or not.

8.2 Patients will have a pre-treatment briefing undertaken which will include administration technique and potential side effects and remedial therapy.

8.3 Patients will sign a consent form.

8.4 T-VEC will be prescribed on the ChemoCare database, and the patient booked for a 1 hour slot in T-VEC clinic.

8.5 T-VEC will be ordered by pharmacy according to the amount prescribed,

8.6 Operators who will administer T-VEC are currently Dr Scatchard, Sally Tapp Lead Immunotherapy CNS and and Immunotherapy Clinical Nurse Specialists who have undergone training.

8.7 T-VEC is provided in single-use vials of 1 mL each in two different dose strengths:

a) 10^6 (1 million) plaque-forming units (PFU) per mL (light green cap) – for initial dose only.
b) 10^8 (100 million) PFU per mL (royal blue cap) – for all subsequent doses

8.8 The total injection volume for each treatment visit should not exceed 4 mL for all injected lesions combined. It may not be possible to inject all lesions at each treatment visit or over the full course of treatment. Previously injected and/or uninjected lesion(s) may be injected at subsequent treatment visits.

8.9 The initial recommended dose is up to 4 mL of T-VEC at a concentration of 10^6 (1 million) PFU per mL. The recommended dose for subsequent administrations is up to 4 mL of T-VEC at a concentration of 10^8 (100 million) PFU per mL.

8.10 Up to 10 lesions will be measured and photographed.

8.11 T-VEC will be stored in -80 freezer and unthawed over 30 mins and labelled, once the patient has arrived in the hospital and the ChemoCare prescription has been

authorised and clinically checked. The administrator will be contacted when it is ready to be collected.

8.12 Time T-VEC taken out of freezer and administration time will both be recorded on the treatment record sheet by the pharmacy.

8.13 Protective clothing including protective goggles, gown and gloves will be worn. Avoid any accidental exposure especially contact with skin/eyes and mucous membranes. Cover any wounds before handling.

8.14 Perform hand hygiene, open sterile dressing pack, then drawn up into the syringe with 22-26 G needle. N.B. each vial contains 1ml, so ensure sufficient has been ordered. Change needle according to size of lesion to be injected. Remove gloves.

8.15 Perform hand hygiene.

8.16 Local anaesthetic is unlikely to be required as this would need additional needle punctures. If amitop has been applied, wipe off before preparing skin for injection.

8.17 Swab with ChloraPrep 2%w/v/70%v/v Cutaneous Solution or Sani-Cloth CHG 2% and allow to dry for 15 seconds.

8.18 Inject lesion(s) according to dose and administration schedule. Needle to be changed after each lesion.

8.19 Gloves to be changed after injecting ALL lesions and before dressing is applied.

8.20 Injected lesions will be covered with a clear film wound dressing and the external surface of the dressing wiped with an alcohol swab. Dressings should be kept on ideally for 8 days or until lesion has dried up.

8.21 Complete treatment administration record and update the patient's alert card with treatment date and batch number.

8.22 Clean all surfaces that may have come into contact with T-VEC, and treat all spills, with a virucidal agent – a 1000ppm solution of Chlorclean or Clorox wipes.

8.23 Dispose of all materials that may have come in contact with T-VEC in a Wiva type 30 litre clinical waste container. Which should then be placed in the disposal cupboard for collection

8.24 In the event of an accidental occupational exposure (e.g. through a splash to the eyes or mucous membranes), flush with clean water for at least 15 minutes.

8.25 In the event of exposure to broken skin or needle stick, clean the affected area thoroughly with soap and water and/or disinfectant.

8.26 Following first treatment the Immunotherapy CNS will inform the patient's GP and Community Nurse, if appropriate.

9. ADMISSION OF PATIENT RECEIVING T-VEC

9.1 Isolation of a patient with Herpes Simplex Virus is usually only necessary if the patient has extensive infection e.g. eczema herpeticum.

9.2 The risk to other patients would be dependent on the vulnerability of the other patients. So if admitted to Yeo/Yarty the patient should be nursed in a single room but otherwise should be able to be managed in a bay as long as occlusive dressings in place.

9.3 The most important infection control measure is to use gloves when dealing with lesions, secretions or bodily fluids and hand hygiene must be undertaken after glove removal.

9.4 Precautions apply to patients admitted within 30 days of a T-VEC injection

10. ARCHIVING ARRANGEMENTS

The original of this SOP will remain with the author Sally Tapp, Immunotherapy CNS Cancer Services. An electronic copy will be maintained on the Trust intranet, C – Cherrybrook, Y – Yeo and Yarty & Yarty Day Case. Archived electronic copies will be stored on the Trust's “archived policies” shared drive, and will be held indefinitely. A paper copy (where one exists) will be retained for 10 years

11. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE STANDARD OPERATING PROCEDURE

11.1 To evidence compliance with this policy, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
Compliance against prescribing	Review of prescriptions against authorised prescribers	Annually at chemotherapy governance
Compliance against administering	Spot audit	Annually at chemotherapy governance

12. REFERENCES

Talimogene laherparepvec for treating unresectable metastatic melanoma. NICE Technology appraisal guidance [TA410] Published date: 28 September 2016.

Andtbacka, R. H. et al. 2015. 'Talimogene Laherparepvec Improves Durable Response Rate in Patients with Advanced Melanoma', *J Clin Oncol*, 33: 2780-8.

13. ACKNOWLEDGEMENTS

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APPENDIX 1: RECOMMENDED DOSE AND SCHEDULE FOR T-VEC ADMINISTRATION

Treatment	Treatment Interval	Maximum Injection Volume per Treatment Visit (all lesions combined)	Dose Strength	Prioritization of Lesions to be Injected
Initial	–	4 mL	10^6 (1 million) PFU per mL	• Inject largest lesion(s) first. • Prioritize injection of remaining lesion(s) based on lesion size until maximum injection volume is reached or until all injectable lesion(s) have been treated.
Second	3 weeks after initial treatment	4 mL	10^8 (100 million) PFU per mL	• Inject any new lesion(s) (lesions that have developed since initial treatment) first. • Prioritize injection of remaining lesion(s) based on lesion size until maximum injection volume is reached or until all injectable lesion(s) have been treated.
All subsequent treatments (including reinitiation)	2 weeks after previous treatment	4 mL	10^8 (100 million) PFU per mL	• Inject any new lesion(s) (lesions that have developed since previous treatment) first. • Prioritize injection of remaining lesion(s) based on lesion size until maximum injection volume is reached or until all injectable lesion(s) have been treated.

Determination of T-VEC injection volume based on lesion size

Lesion Size (longest dimension)	Injection Volume
> 5 cm	up to 4 mL
> 2.5 cm to 5 cm	up to 2 mL
> 1.5 cm to 2.5 cm	up to 1 mL
> 0.5 cm to 1.5 cm	up to 0.5 mL
≤ 0.5 cm	up to 0.1 mL

When lesions are clustered together, inject them as a single lesion according to above. Continue T-VEC treatment for at least 6 months unless other treatment is required or until there are no injectable lesions to treat.

Reinitiate T-VEC treatment if new unresectable cutaneous, subcutaneous, or nodal lesions appear after a complete response.

APPENDIX 2: CLOSE CONTACTS INFORMATION

INFORMATION FOR PATIENTS AND CLOSE CONTACTS

This brochure contains important safety information you should know before and during IMLYGIC® treatment.
Please read this information carefully before and after each treatment.
Please also read the IMLYGIC® Package Leaflet that will be handed to you by your healthcare professional and is also available online at http://www.medicines.org.uk/emc/

Talk to your doctor or nurse if you have any questions or concerns about your treatment.

WHAT SHOULD I KNOW ABOUT IMLYGIC®?

- IMLYGIC® is a weakened form of herpes simplex virus type 1 (HSV-1), which is commonly called the cold sore virus.
- IMLYGIC® may cause life-threatening herpes infection in persons with weakened immune systems.
- IMLYGIC® may harm your unborn baby. Tell your doctor if you are pregnant or plan to become pregnant.
- During or after IMLYGIC® treatment, you could develop cold sores or a more serious herpes infection.
- IMLYGIC® virus could be spread to your close contacts (household members, caregivers, sex partners, or someone you share a bed with) if they have direct contact with your body fluids or injection sites.
- You should share this information with your close contacts, particularly if they are pregnant or have a weakened immune system.
- Injected sites should be covered with airtight and watertight dressings at all times. Apply the dressing as instructed by your healthcare professional and ask your healthcare professional how long the injection site should be covered for. If the dressing comes loose or falls off, replace it right away with a clean dressing.

WHAT ARE SIGNS AND SYMPTOMS OF HERPES INFECTION?

- Pain, burning or tingling in a blister around the mouth or genitals (e.g. cold sore), or on the finger or ears with or without a scab
- Eye pain, light sensitivity, discharge from the eyes, redness of eye, tearing, or blurry vision
- Weakness in arms or legs
- Extreme drowsiness (feeling sleepy)
- Fever associated with headache
- Vomiting
- Mental confusion
- Memory loss
- Seizures

IF YOU HAVE ANY OF THESE SYMPTOMS DURING OR AFTER TREATMENT WITH IMLYGIC®, TELL YOUR DOCTOR RIGHT AWAY.

Please continue to reverse side for additional information.

This sheet does not include all the important safety information you need to know about IMLYGIC®. Please refer to the IMLYGIC® Package Leaflet that will be handed to you by your healthcare professional and is also available online at http://www.medicines.org.uk/emc/ for more information. You should read this at the time of each injection because the information may change over time.

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